

Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

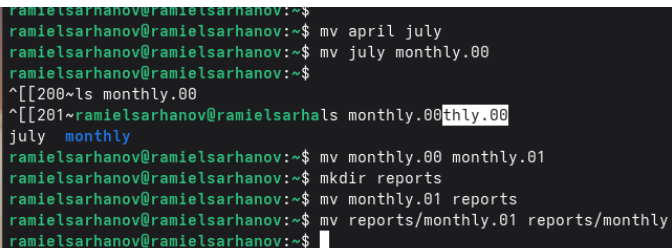
- 1 Выполнить приимеры
- 2 Выполнить дествия по работе с каталогами и файлами
- 3 Выполнить действия с правами доступа
- 4 Получить дополнительные сведения при помощи справки по командам.

2. Процесс выполнения лабораторной работы

Выполнение примеров

```
ramielsarhanov@ramielsarhanov:~$ touch abc1
ramielsarhanov@ramielsarhanov:~$ cp abc1 april
ramielsarhanov@ramielsarhanov:~$ cp abc1 may
ramielsarhanov@ramielsarhanov:~$ ^C
ramielsarhanov@ramielsarhanov:~$
^[[200~cp abc1 may
^[[201~ramielsarhanov@ramielsarhanov:~$ cp abc1 may
ramielsarhanov@ramielsarhanov:~$ mkdir monthly
ramielsarhanov@ramielsarhanov:~$ cp april may monthly
ramielsarhanov@ramielsarhanov:~$ cp monthly/may monthly/june
ramielsarhanov@ramielsarhanov:~$ ls monthly
april  june  may
ramielsarhanov@ramielsarhanov:~$
^[[200~mkdir monthly.00
^[[201~ramielsarhanov@ramielsarhanov:~$ mkdir monthly.00
ramielsarhanov@ramielsarhanov:~$ cp -r monthly monthly.00
ramielsarhanov@ramielsarhanov:~$ cp -r monthly.00 /tmp
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 1: Выполнение примеров



```
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ mv april july  
ramielsarhanov@ramielsarhanov:~$ mv july monthly.00  
ramielsarhanov@ramielsarhanov:~$  
^[[200~ls monthly.00  
^[[201~ramielsarhanov@ramielsarhanov:~$ ls monthly.00  
july  monthly  
ramielsarhanov@ramielsarhanov:~$ mv monthly.00 monthly.01  
ramielsarhanov@ramielsarhanov:~$ mkdir reports  
ramielsarhanov@ramielsarhanov:~$ mv monthly.01 reports  
ramielsarhanov@ramielsarhanov:~$ mv reports/monthly.01 reports/monthly  
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 2: Выполнение примеров


```
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ touch may  
ramielsarhanov@ramielsarhanov:~$ ls -l may  
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:05 may  
ramielsarhanov@ramielsarhanov:~$ chmod u+x may  
ramielsarhanov@ramielsarhanov:~$ ls -l may  
-rwxr--r--. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:05 may  
ramielsarhanov@ramielsarhanov:~$ chmod u-x may  
ramielsarhanov@ramielsarhanov:~$ ls -l may  
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:05 may  
ramielsarhanov@ramielsarhanov:~$ chmod g-r,o-r monthly  
ramielsarhanov@ramielsarhanov:~$ chmod g+w abc1  
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 3: Выполнение примеров

Создание директорий и копирование файлов

```
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ cp /usr/include/linux/sysinfo.h ~  
ramielsarhanov@ramielsarhanov:~$ mv sysinfo.h equipment  
ramielsarhanov@ramielsarhanov:~$ mkdir ski.plases  
ramielsarhanov@ramielsarhanov:~$ mv equipment ski.plases/  
ramielsarhanov@ramielsarhanov:~$ mv ski.plases/equipment ski.plases/equiplist  
ramielsarhanov@ramielsarhanov:~$ touch abc1  
ramielsarhanov@ramielsarhanov:~$ cp abc1 ski.plases/equiplist2  
ramielsarhanov@ramielsarhanov:~$ cd ski.plases/  
ramielsarhanov@ramielsarhanov:~/ski.plases$ mkdir equipment  
ramielsarhanov@ramielsarhanov:~/ski.plases$ mv equiplist equipment/  
ramielsarhanov@ramielsarhanov:~/ski.plases$ mv equiplist2 equipment/  
ramielsarhanov@ramielsarhanov:~/ski.plases$ cd  
ramielsarhanov@ramielsarhanov:~$ mkdir newdir  
ramielsarhanov@ramielsarhanov:~$ mv newdir ski.plases/  
ramielsarhanov@ramielsarhanov:~$ mv ski.plases/newdir/ ski.plases/plans  
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 4: Работа с каталогами

Работа с командой chmod

```
ramielsarhanov@ramielsarhanov:~$ mkdir australia play
ramielsarhanov@ramielsarhanov:~$ touch my_os feathers
ramielsarhanov@ramielsarhanov:~$ chmod 744 australia/
ramielsarhanov@ramielsarhanov:~$ chmod 711 play/
ramielsarhanov@ramielsarhanov:~$ chmod 544 my_os
ramielsarhanov@ramielsarhanov:~$ chmod 664 feathers
ramielsarhanov@ramielsarhanov:~$ ls -l
total 0
-rw-rw-r--. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:06 abc1
drwxr--r--. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:09 australia
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 15:08 Desktop
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 15:08 Documents
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 15:08 Downloads
-rw-rw-r--. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:09 feathers
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 74 Sep  2 15:41 git-extended
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:05 may
drwx--x--x. 1 ramielsarhanov ramielsarhanov 24 Sep  2 16:02 monthly
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 15:08 Music
-r-xr--r--. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:09 my_os
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 15:08 Pictures
drwx--x--x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 16:09 play
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 15:08 Public
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 14 Sep  2 16:03 reports
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 28 Sep  2 16:06 ski.places
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 15:08 Templates
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 0 Sep  2 15:08 Videos
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov 10 Sep  2 15:19 work
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 5: Настройка прав доступа

```
ramielsarhanov@ramielsarhanov:~ — less /etc/passwd
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/:usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:usr/sbin/nologin
polkitd:x:114:114>User for polkitd:/:sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/:usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
openvpn:x:990:990:OpenVPN:/etc/openvpn:/usr/sbin/nologin
nm-openvpn:x:989:989:Default user for running openvpn spawned by NetworkManager:/:usr/sbin/nologin
abrt:x:173:173:/:etc/abrt:/usr/sbin/nologin
wsdd:x:987:987:Web Services Dynamic Discovery host daemon:/:usr/sbin/nologin
nm-openconnect:x:986:986:NetworkManager user for OpenConnect:/:usr/sbin/nologin
rtkit:x:985:985:RealtimeKit:/:usr/sbin/nologin
/etc/passwd
```

```
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ cp feathers file.old  
ramielsarhanov@ramielsarhanov:~$ mv file.old play/  
ramielsarhanov@ramielsarhanov:~$ mkdir fun  
ramielsarhanov@ramielsarhanov:~$ cp -R play/ fun/  
ramielsarhanov@ramielsarhanov:~$ mv fun/ play/games  
ramielsarhanov@ramielsarhanov:~$ chmod u-r feathers  
ramielsarhanov@ramielsarhanov:~$ cat feathers  
cat: feathers: Permission denied  
ramielsarhanov@ramielsarhanov:~$ cp feathers feathers2  
cp: cannot open 'feathers' for reading: Permission denied  
ramielsarhanov@ramielsarhanov:~$ chmod u+r feathers  
ramielsarhanov@ramielsarhanov:~$ chmod u-x play/  
ramielsarhanov@ramielsarhanov:~$ cd play/  
bash: cd: play/: Permission denied  
ramielsarhanov@ramielsarhanov:~$ chmod +x play/  
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 7: Работа с файлами и правами доступа

```
MOUNT(8)                                     System Administration                                     MOUNT(8)
```

NAME

mount - mount a filesystem

SYNOPSIS

mount [-h|-V]

mount [-l] [-t *fstype*]

mount -a [-fFnrsvw] [-t *fstype*] [-O *optlist*]

mount [-fnrsvw] [-o *options*] *device*|*mountpoint*

mount [-fnrsvw] [-t *fstype*] [-o *options*] *device* *mountpoint*

mount --bind|--rbind|--move *olddir* *newdir*

mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] *mountpoint*

DESCRIPTION

All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at `/`. These files can be spread out over several devices. The `mount` command serves to attach the filesystem found on some device to the big file tree. Conversely, the `umount(8)` command will detach it again. The filesystem is used to control how data is stored on the device or provided in a virtual way by network or other services.

The standard form of the `mount` command is:

`mount -t type device dir`

This tells the kernel to attach the filesystem found on *device* (which is of type *type*) at the directory *dir*. The option `-t type` is optional. The `mount` command is usually able to detect a filesystem. The root permissions are necessary to mount a filesystem by default. See section "Non-superuser mounts" below for more details. The previous contents (if any) and owner and mode of

Manual page mount(8) line 1 (press h for help or q to quit)

```
FSCK(8)                                System Administration                                FSCK(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be a device
    name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem label or UUID
    specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root). Normally, the fsck program
    will try to handle filesystems on different physical disk drives in parallel to reduce the total
    amount of time needed to check all of them.

    If no filesystems are specified on the command line, and the -A option is not specified, fsck will
    default to checking filesystems in /etc/fstab serially. This is equivalent to the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0      No errors

    1      Filesystem errors corrected

    2      System should be rebooted

    4      Filesystem errors left uncorrected

    8      Operational error

Manual page fsck(8) line 1 (press h for help or q to quit)
```

```

MKFS(8)                                     System Administration                               MKFS(8)

NAME
    mkfs - build a Linux filesystem

SYNOPSIS
    mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
    This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

    mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device
    argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall contain
    the filesystem. The size argument is the number of blocks to be used for the filesystem.

    The exit status returned by mkfs is 0 on success and 1 on failure.

    In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.fstype) available
    under Linux. The filesystem-specific builder is searched for via your PATH environment setting only.
    Please see the filesystem-specific builder manual pages for further details.

OPTIONS
    -t, --type type
        Specify the type of filesystem to be built. If not specified, the default filesystem type
        (currently ext2) is used.

    fs-options
        Filesystem-specific options to be passed to the real filesystem builder.

    -V, --verbose
        Produce verbose output, including all filesystem-specific commands that are executed. Specifying
        this option more than once inhibits execution of any filesystem-specific commands. This is really
        only useful for testing.

    -h, --help
        Manual page mkfs(8) line 1 (press h for help or q to quit)

```



```
KILL(1)                                User Commands                                KILL(1)

NAME
    kill - terminate a process

SYNOPSIS
    kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds signal] [--] pid|name...

    kill -l [number|0xsigmask] | -L

    kill -d pid

DESCRIPTION
    The command kill sends the specified signal to the specified processes or process groups.

    If no signal is specified, the TERM signal is sent. The default action for this signal is to terminate the process. This signal should be used in preference to the KILL signal (number 9), since a process may install a handler for the TERM signal in order to perform clean-up steps before terminating in an orderly fashion. If a process does not terminate after a TERM signal has been sent, then the KILL signal may be used; be aware that the latter signal cannot be caught, and so does not give the target process the opportunity to perform any clean-up before terminating.

    Most modern shells have a builtin kill command, with a usage rather similar to that of the command described here. The --all, --pid, and --queue options, and the possibility to specify processes by command name, are local extensions.

    If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
    The list of processes to be signaled can be a mixture of names and PIDs.

    pid
        Each pid can be expressed in one of the following ways:

        n

Manual page kill(1) line 1 (press h for help or q to quit)
```

3. Выводы по проделанной работе

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.